Print table - LOOPING B3

Print the multiplication table as per the multiplicant and n.

Input Format

Integer Integer

Constraints

NULL

Output Format

Integer * Integer = Integer

Sample Input 0

5 4

Sample Output 0

1 * 5 = 5
2 * 5 = 10
3 * 5 = 15
4 * 5 = 20

```
import java.io.; import java.util.; import java.text.; import java.math.; import java.util.regex.*;

public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should
be named Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int t=sc.nextInt();
    for(int i=1;i<=t;i++){
        System.out.println(i+" * "+n+" = "+(n*i));
    }
}
}
```

Fibonacci Sequence 24

Display Fibonacci Sequence , from the given number The Fibonacci sequence is a sequence where the next term is the sum of the previous two terms. The first two terms of the Fibonacci sequence are 0 followed by 1.

Input Format

INT

Constraints

NULL

Output Format

N Fibonacci Sequence

Sample Input 0

10

Sample Output 0

0, 1, 1, 2, 3, 5, 8, 13, 21, 34,

Sample Input 1

8

Sample Output 1

0, 1, 1, 2, 3, 5, 8, 13,

```
import java.io.; import java.util.; import java.text.; import java.math.; import java.util.regex.*;

public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should
be named Solution. */
    Scanner sc=new Scanner(System.in);
```

```
    int n=sc.nextInt();
    int a=0,b=1;
    System.out.print("0, 1, ");
    for(int i=3;i<=n;i++){
        int c=a+b;
        System.out.print(c+", ");
        a=b;
        b=c;
    }
}
 }
```

---

factors Loop S3

Print all the factors of the given number.

Input Format

INT

Constraints

NULL

Output Format

Set of integers seperated by comma.

Sample Input 0

12

Sample Output 0

1,2,3,4,6,12

Explanation 0

Factors of 12 = 1,2,3,4,6,12

import java.io.*; import java.util.*; import java.text.*; import java.math.*; import java.util.regex.*;

```
public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    for(int i=1;i<=n;i++){
        if(n%i==0){
            if(i!=n)
                System.out.print(i+",");
            else
                System.out.print(i);
        }
    }
}
 }
```

Count and Sum of Factors of a Number

Given an integer N, write a program to:

Count the total number of factors of the number.

Find the sum of all its factors.

A factor of a number is an integer that divides the number exactly without leaving a remainder.

Input Format

The first line contains an integer N.

Constraints

Nil

Output Format

Print two integers separated by a space:

First → total number of factors

Second → sum of all factors

 Explanation Factors of 6 → 1, 2, 3, 6 Count = 4 Sum = 12

Sample Input 0

6

Sample Output 0

4 12

Sample Input 1

10

Sample Output 1

4 18

import java.io.*; import java.util.*; import java.text.*; import java.math.*; import java.util.regex.*;

public class Solution {

```
public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int sum=0,count=0;
    for(int i=1;i<=n;i++){
        if(n%i==0){
            count++;
            sum+=i;
        }
    }
    System.out.println(count+" "+sum);
}
 }
```

PAD

Given a number, check whether it is an abundant or deficient or perfect number N number 6 , find the factors and add it eg 6 = 1 , 2 , 3

the given number and sum , if both are same = Perfect Number

the given number lesser than sum = Abundant Number

the given number greater than sum = Deficient Number

Input Format

Integer

Constraints

1 < n < 100000

Output Format

Abundant/Decient/Perfect Number

Sample Input 0

6

Sample Output 0

Perfect Number

Explanation 0

6 = sum of the factor 1 + 2 + 3 = 6

Sample Input 1

7

Sample Output 1

Deficient Number

import java.io.*; import java.util.*; import java.text.*; import java.math.*; import java.util.regex.*;

```
public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int sum=0;
    for(int i=1;i<=n/2;i++){
        if(n%i==0){
            sum+=i;
        }
    }
    System.out.println(sum==n?"Perfect Number":sum>n?"Abundant Number":"Deficient Number");
}
 }
```

Prime or Not 44

Check Whether a Number is Prime or Not

Input Format

INT

Constraints

1 <= n <= 10^20

Output Format

__ is a prime number. __ is not a prime number.

Sample Input 0

5

Sample Output 0

5 is a prime number.

Sample Input 1

10

Sample Output 1

10 is not a prime number.

```
import java.io.; import java.util.; import java.text.; import java.math.; import java.util.regex.*;

public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should
be named Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int count=0;
    for(int i=1;i<=n;i++){
        if(n%i==0){
            count++;
        }
    }
    if(count==2)
        System.out.println(n+" is a prime number.");
    else
        System.out.println(n+" is not a prime number.");
}
 }
```

Amicable pair S3

Find whether the given number is amicable pair or not.(sum of factors of first number equals to the 2nd number as well as sum of factors of 2nd number equal to first number)

Input Format

INT INT

Constraints

NULL

Output Format

Amicable Pair or Not a Amicable Pair

Sample Input 0

220 284

Sample Output 0

Amicable Pair

Sample Input 1

123 456

Sample Output 1

Not a Amicable Pair

```
import java.io.; import java.util.; import java.text.; import java.math.; import java.util.regex.*;

public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named
Solution. */
    Scanner sc=new Scanner(System.in);
    int num1=sc.nextInt();
    int num2=sc.nextInt();
    int sum1=0,sum2=0;
    for(int i=1;i<=num1/2;i++){
        if(num1%i==0){
            sum1+=i;
        }
    }
    for(int j=1;j<=num2/2;j++){
        if(num2%j==0){
            sum2+=j;
        }
    }
    System.out.println(sum1==num2&&sum2==num1?"Amicable Pair":"Not a Amicable Pair");
}
 }
```

LCM code

Find the LCM of the given two numbers.

Input Format

Integer Integer

Constraints

null

Output Format

Integer

Sample Input 0

5 8

Sample Output 0

40

```
import java.io.; import java.util.; import java.text.; import java.math.; import java.util.regex.*;

public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should
be named Solution. */
    Scanner sc=new Scanner(System.in);
    int a=sc.nextInt();
    int b=sc.nextInt();
    int big=0,ans=0;
    if(a>b)
        big=a;
    else
        big=b;
    for(int i=1;i<=big;i++){
        if(a%i==0 && b%i==0){
```

```
            ans=i;
        }
    }
    System.out.println((a*b)/ans);
}
 }
```

---

perfect square 26

Check whether given number is perfect square or not.

Input Format

INT

Constraints

NULL

Output Format

Perfect Square / Not

Sample Input 0

10000

Sample Output 0

Perfect Square

Sample Input 1

145

Sample Output 1

Not

import java.io.*; import java.util.*; import java.text.*; import java.math.*; import java.util.regex.*;

public class Solution {

```
public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    boolean b=false;
    for(int i=2;i<=n;i++){
        if(i*i==n){
            System.out.println("Perfect Square");
            b=true;
            break;
        }
    }
    if(b==false){
        System.out.println("Not");
    }
}
 }
```

Square Star Pattern

Program to Print Square Star Pattern

Input Format

Int

Constraints

NULL

Output Format

Square Star Pattern

Sample Input 0

5

Sample Output 0

```
*****
*****
*****
*****
*****
```

Sample Input 1

7

Sample Output 1

```
*******
*******
*******
*******
*******
*******
*******
```

```
import java.io.; import java.util.; import java.text.; import java.math.; import java.util.regex.*;

public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should
be named Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            System.out.print("*");
        }
        System.out.println();
    }
}
 }
```

Rhombus Star Pattern

Print Rhombus Star Pattern

Input Format

INT

Constraints

NULL

Output Format

Pattern

Sample Input 0

4

Sample Output 0

```
****
 ****
  ****
   ****
```

Sample Input 1

6

Sample Output 1

```
******
 ******
  ******
   ******
    ******
     ******
```

import java.io.*; import java.util.*; import java.text.*; import java.math.*; import java.util.regex.*;

public class Solution {

```
public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    for(int i=0;i<n;i++){
        for(int j=0;j<i+n;j++){
            if(j<i)
                System.out.print(" ");
            else
                System.out.print("*");
        }
        System.out.println();
    }
}
 }
```

Right triangle pattern

Right triangle pattern

right angle triangle that has its perpendicular line at the right side of the triangle.

Input Format

int

Constraints

NULL

Output Format

Right angle based on the number

Sample Input 0

5

Sample Output 0

```
    *
   **
  ***
 ****
*****
```

Sample Input 1

3

Sample Output 1

```
  *
 **
***
```

```
import java.io.; import java.util.; import java.text.; import java.math.; import java.util.regex.*;

public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named
Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i+j>=n-1)
                System.out.print("*");
            else
                System.out.print(" ");
        }
        System.out.println();
    }

}
 }
```

Pyramid Star Pattern 3

Print Pyramid Star Pattern

Input Format

INT

Constraints

NULL

Output Format

triangle shape

Sample Input 0

4

Sample Output 0

```
    *
   ***
  *****
*******
```

Sample Input 1

7

Sample Output 1

```
          *
         ***
        *****
       *******
      *********
     ***********
*************
```

import java.io.*; import java.util.*; import java.text.*; import java.math.*; import java.util.regex.*;

public class Solution {

```
public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named
Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int i,j,sp;
    for(i=1;i<=n;i++){
        for(sp=n-1;sp>=i;sp--){
            System.out.print(" ");
        }
        for(j=1;j<i*2;j++){
            System.out.print("*");
        }

        System.out.println();
    }
}
 }
```

Half Diamond Star Pattern 3

Print Half Diamond Star Pattern (odd number only will be given)

Input Format

Int (odd number)

Constraints

NULL

Output Format

pattern

Sample Input 0

5

Sample Output 0

```
*
**
***
**
*
```

Sample Input 1

7

Sample Output 1

```
*
**
***
****
***
**
*
```

import java.io.*; import java.util.*; import java.text.*; import java.math.*; import java.util.regex.*;

```
public class Solution {

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named
Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    for(int i=0;i<n;i++){
        for(int j=0;j<=i;j++){
            if(i+j>=n)
                System.out.print(" ");
            else
                System.out.print("*");
        }
        System.out.println();
    }

}
 }
```

Inverted Half Diamond Star Pattern

Print Inverted Half Diamond Star Pattern

Input Format

INT (odd number)

Constraints

NULL

Output Format

Pattern

Sample Input 0

3

Sample Output 0

```
  *
**
  *
```

Sample Input 1

5

Sample Output 1

```
   *
  **
***
  **
   *
```

import java.io.*; import java.util.*; import java.text.*; import java.math.*; import java.util.regex.*;

public class Solution {

```
public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named
Solution. */
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int i,j,sp;
    for(i=0;i<=n/2;i++){
        for(sp=n/2;sp>i;sp--){
            System.out.print(" ");
        }
        for(j=0;j<=i;j++){
            System.out.print("*");
        }
        System.out.println();
    }
    for(i=0;i<n/2;i++){
        for(sp=0;sp<=i;sp++){
            System.out.print(" ");
        }
        for(j=n/2;j>i;j--){
            System.out.print("*");
        }
        System.out.println();
    }
}
```